



Thermal conductivity of Li-ion batteries and their electrode configurations – A novel combination of modelling and experimental approach



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HIGHLIGHTS

- Unit cell thermal modelling of stacked electrodes.
- Thermal conductivity model for electrode stacks and jelly rolls.
- Temperature dependent material properties.
- Correlations for application in thermal models.

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ABSTRACT

A bottom-up approach to calculate the overall and averaged thermal properties of the jelly roll or electrode stack of Li-ion cells in a generally applicable way is introduced. The model is based on temperature-dependent material properties and is specifically applied to a prismatic hardcase cell. The geometrical properties, such as width, depth, height, porosity, tortuosity, and the saturation with electrolyte are considered. Most of the material properties are determined by own measurements within a wide temperature range and compared with literature data. Anisotropic unit cell properties for homogenised three-dimensional thermal models and the corresponding representative thermal conductivity for one- or two-dimensional models can be calculated. A non-destructive measurement technique to determine the thermal conductivity of prismatic hardcase cell geometries and validate the described model is presented.

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1. Introduction

The automotive application of Li-ion batteries as power source for (hybrid) electric vehicles requires a thermal management system to maintain performance and ensure a safe and harmless operation under various thermal boundary conditions [1,2]. High power and high energy automotive cells exhibit a non-uniform internal temperature distribution mainly due to their operational heat losses and large-format dimensions, respectively. Furthermore, the electrochemical reactions and the cell components' material properties are strongly dependent on temperature. The core temperature as input parameter for a reliable management

system is significantly different from the surface temperature [3,4] and has to be deduced from electrical signals [5,6] or calculated based on thermal models.

In order to develop accurate numerical simulations, the underlying material properties are of critical importance [7]. Several non-destructive top-down approaches have been published in literature to identify the predominant thermal properties of Li-ion cells such as overall thermal conductivity. These investigations mainly focus on cylindrical [8–10] or pouch-bag cells [11–16] and mostly neglect the temperature dependency. Bazinski et al. did not discover a significant influence of temperature on the overall thermal conductivity of a pouch-bag cell [16], which is in contrast to numerous material properties [17–19]. A dependency on state of charge has been identified by different authors [8,12,16].

The non-destructive characterisation methodologies are beneficial for lumped thermal models of reduced complexity, but impede a thoroughly investigation of the individual layers and

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geometry inside the cells and their contribution to the overall thermal conductivity. However, the experimental methodologies developed for cylindrical and pouch-bag cells are not applicable to prismatic hardcase cells as they are used in current automotive applications.

An extensive study on the effective thermal conductivity of electrodes has been presented by Vadakkepatt et al. [20]. The authors investigated several effective medium theories and compared these results to the effective thermal conductivity obtained from thermal simulations on a reconstructed cathode microstructure. The numerical approach links the strong influence of structure porosity to different thermal conductivity ratios inside the solid phase. The authors state, that the few data available in literature is not suitable for the validation of the numerical results, due to just partially documented physical data of the belonging materials.

In this work the connection of thermal conductivity of electrode coatings, their configuration as electrode stacks and the overall thermal transport behaviour of resulting geometric arrangements such as jelly rolls are investigated. This is the first time a consistent approach is presented, that spans from single material thermal modelling to experimental validation at cell level of hardcase cells. Four mayor aspects are addressed:

- (i) A thermal conductivity model for each porous layer (anode, cathode and separator) based on own measurements of thermal material properties that cover a wide temperature range applicable for automotive applications.
- (ii) A profound thermal conductivity model ranging from anisotropic electrode stack unit cells to more specific thermal transport behaviour of jelly roll geometries at cell level.
- (iii) Experimental methodology for non-destructive determination of the overall thermal conductivity of prismatic hardcase cells.
- (iv) Validation of the thermal conductivity models by comparison with experimental results.

As a bottom-up approach, the thermal modelling is based on temperature-dependent material properties of each unit cell layer, namely the electrolyte saturated separator, cathode and anode coatings as well as the current collectors. Each layer is described accurately regarding width, depth, height, porosity and the saturation with electrolyte. The complete modelling approach from porous media thermal properties to the overall jelly roll thermal transport behaviour is combined into the Thermal Resistance Model (TRM) that is described in detail in Chapter 4.1.

The remainder of the paper is organized in four chapters. The theory and modelling chapter explains the related theory in terms of thermal material properties as well as the modelling approach which is used to evaluate the experiments. In the experimental chapter, the developed set-up and the prismatic hardcase cell experiments are presented. The results are analysed and discussed in the corresponding results chapter, which is followed up by the conclusions.

2. Theory and modelling approach

2.1. Thermal properties

A material's transient or steady state thermal characteristics can be described by its thermal diffusivity a , which is dependent on temperature and pressure. The thermal diffusivity is defined as the quotient of the thermal conductivity k , the gravimetric density ρ and the specific heat capacity c_p of the considered material.

$$a(T, p) = \frac{k}{\rho \cdot c_p} \quad (1)$$

The dependency on pressure is often negligible for solid and liquid materials [21]. All materials that are integrated in Li-ion cells can therefore be treated as solely temperature dependent for this investigation.

If different materials are coupled in serial or parallel direction, the resulting thermal conductivities $k_{\perp}$ and $k_{\parallel}$ for heat flows can be described as follows:

$$\frac{k_{\perp}}{l_c} = \left(\sum_i^I \left(\frac{k_i}{s_i} \right)^{-1} \right)^{-1} \quad (2)$$

$$\frac{k_{\parallel}}{l_c} = \sum_i^I \frac{k_i}{s_i} \quad (3)$$

Based on the geometric arrangement of the materials $i = 1 \dots I$, their individual thermal conductivity k_i and dimension s_i , an overall thermal conductivity in terms of a characteristic length l_c can be calculated using Equations (2) and (3) for series and parallel connection, respectively. Furthermore, a specific thermal resistance R_{th} can be determined with the cross-sectional area A , see Equation (4).

$$R_{th} = \frac{l_c}{k \cdot A} \quad (4)$$

Thermal resistances can be utilised in thermal equivalent circuits according to the electro-thermal analogy [22].

2.2. Porous media

Thermal properties of materials that are denoted in literature usually refer to a homogeneous bulk material. The coated electrode materials of the anode and cathode as well as the separator however, show a significant volumetric porosity ranging from 30% to 50% [17,23] that is filled with electrolyte. Consequently, the thermal conductivity of the two-phase porous components has to be modelled by an adequate combination of the solid and the liquid phase. A commonly used approach is the Krischer model [19,24] which is based on the weighted series combination of a series and a parallel connection of thermal conductivities: In can be expressed by the following set of Equations in which f_a is the weighting factor and Ψ is the solid phase porosity.

$$k_{two-phase} = \left(\frac{f_a}{k_{series}} + \frac{f_a - 1}{k_{parallel}} \right)^{-1} \quad (5)$$

$$k_{series} = \left(\frac{\Psi}{k_{fluid}} + \frac{1 - \Psi}{k_{solid}} \right)^{-1} \quad (6)$$

$$k_{parallel} = \Psi \cdot k_{fluid} + (1 - \Psi) \cdot k_{solid} \quad (7)$$

The advantage of the Krischer model lies in its scalability of the parallel and the series components that can be adapted to different structures, covering the whole range of porosity without being limited to certain dilutions. Originally the model was developed for packed beds of spherical particles filled with fluid. The weighting factor depicts a kind of tortuosity and was experimentally determined as $f_a = 0.2$ for packed beds [24]. Fig. 1 visualises the interrelation the Equations of the Krischer model.

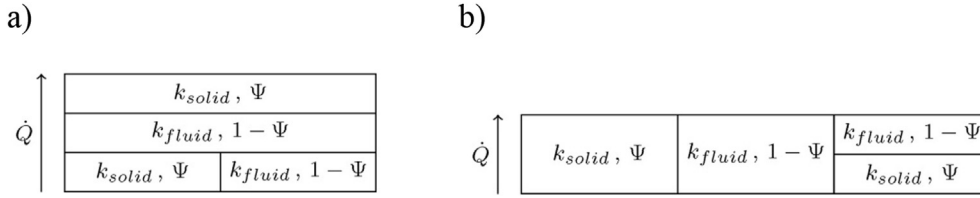


Fig. 1. Thermal conductivity model for fluid-solid-systems in a weighted combination according to a) Krischer and b) adapted Krischer model with a continuous solid phase.

2.2.1. Electrode coatings

Electrical models usually approximate the particles in anode and cathode electrode coatings as spherical particles [25,26]. The model described by Krischer can be used to calculate the thermal conductivity of the electrolyte saturated components inside Li-ion cells based on bulk material data. However, a significant difference between packed beds and the cell components is the presence of binder and carbon black [27] that leads to a continuous solid phase in case of the cell components. The continuous solid phase of the electrode coatings is necessary to maintain the migration of electrons from the current collectors to the intercalating Li-ions and vice versa. To take this effect into account it is reasonable to adapt the porous media model by Krischer in a way that the series and parallel connections of the two phases are combined in parallel set-up as shown in Fig. 1b.

The effectivity and suitability of such transition from series to parallel combination for a physically similar situation has been proven for heat transfer in open-cell foams with a continuous solid material matrix [28].

2.2.2. Separator

A continuous fluid phase with preferably low tortuosity is indispensable to achieve a straight Li-ion diffusion perpendicular through the separator's solid matrix. Due to the characteristic structures of polymer materials and the resulting tortuosity, a straight parallel connection of the electrolyte and polymer phase represents the limiting case of ideal diffusion paths. Several models are proposed in literature to calculate the thermal conductivity of porous polymer structures in case of intrinsic pores. An overview is given by Progelhof et al. [29]. These intrinsic pore models are in fact based on an opposite assumption of the straight parallel connected two phases and resemble the original Krischer series connection. The considerations of intrinsic pores and the series connection however, are in absolute contrast to the requirement of available diffusion paths through the separator. According to these considerations, the authors of this paper propose to use the adapted Krischer model as described above as being suitable for modelling the thermal conductivity of electrolyte saturated separators. The approach will be described in the following paragraphs and the validation will be shown in the results chapter. The weighting factor f_a in Equation (5) has to be redefined for non-spherical continuous solid distributions to consider tortuosity. An ostensive definition of the tortuosity τ is given by Equation (8) in which l_{eff} is the actual diffusion length and d is the direct distance inside the porous media [30].

$$\tau = \left(\frac{l_{eff}}{d} \right)^2 \quad (8)$$

Since tortuosity is difficult to measure for polymer layers in the scale of micrometres, the porosity can be used as the weighting factor f_a for a first approximation.

2.3. Heat transfer

The temperature field equation for conduction heat transfer in solids or stagnant fluids in any coordinate system with internal heat generation is given by Equation (9), in which T is the absolute temperature and $\dot{q}$ is the volume specific heat source.

$$\nabla^2(a \cdot T) + \frac{\dot{q}}{\rho \cdot c_p} = \frac{\partial T}{\partial t} \quad (9)$$

For steady-state and one-dimensional, planar considerations [31] along one coordinate $|x| \in [0, X]$, Equation (9) can be reduced to Equation (10), which is necessary for the one-dimensional thermal model of the cell described in Chapter 2.4.3. The materials properties and the heat source are assumed to be independent on temperature and k is assumed to be isotropic.

$$\frac{d^2 T}{dx^2} + \frac{\dot{q}}{k} = 0 \quad (10)$$

The resulting temperature distribution $T(x)$ can be calculated by means of integration of Equation (10) for a convective boundary condition, see Appendix A for more details.

$$T(x) = T_{fluid} + \frac{\dot{q} \cdot X^2}{2 \cdot k} \cdot \left(1 - \frac{x^2}{X^2} + \frac{2 \cdot k}{ht \cdot c \cdot X} \right) \quad (11)$$

In case of Li-ion cells the specific heat source $\dot{q}$ in Equations (9)–(11) has its origin in the electrochemistry of the electrodes. Both the heat source and the thermal properties of the jelly roll are assumed to be homogeneously distributed over the jelly roll and solely heat conduction is considered. The energy balance of an electrochemical cell can be described by Equation (12), disregarding the enthalpy-of-reaction, enthalpy-of-mixing and phase-change according to [32,33].

$$V \cdot \dot{q}_{el} = I \cdot (U - U_{OCV}) + I \cdot T \cdot \frac{\partial U_{OCV}}{\partial T} + m \cdot c_p \cdot \frac{dT}{dt} \quad (12)$$

In Equation (12) V is the volume of the jelly roll, U is the terminal voltage, U_{OCV} is the cell's open-circuit voltage, I the terminal current, $\dot{q}_{el}$ the volume specific electrical heat source and m the mass of the jelly roll. In case of a symmetric zero root mean square current signal, the second summand can be neglected due to its alternating sign. The integral over a half period of the square-shaped current profile has to be low to reduce the deviation in state of charge.

2.4. Modelling

2.4.1. Unit cell

The jelly roll (JR) is modelled based on the unit cell (UC) shown Fig. 2a. The unit cell, consisting of eight layers, is wound several times around itself, forming the jelly roll, which can be modelled independently in terms of width w , depth d and height h .

The thermal conductivity of each layer of the unit cell is calculated based on the Krischer model or its adaption, respectively.

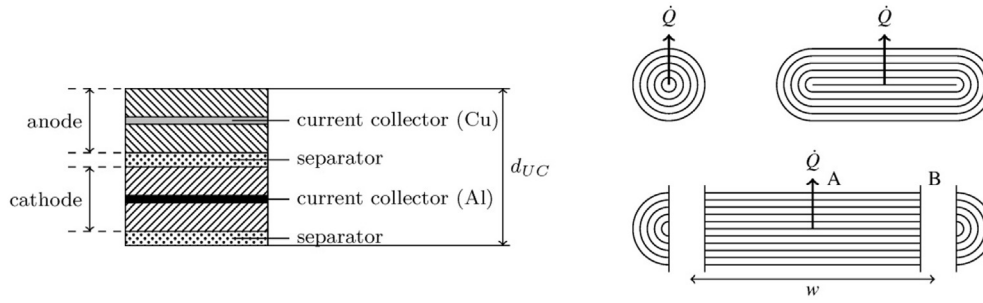


Fig. 2. a) Unit cell consisting of one pair of electrodes and b) three characteristic shapes depending on the width w of the jelly roll and the considered heat flow paths.

Ponnappan et al. [34] stated that the presence of electrolyte remarkably reduces the interfacial resistance between the layers. In their research, a variation of the pressure only showed negligible effect on the wet stack resistance. From that they concluded, that the electrolyte provides a good thermal contact for the electrodes. Furthermore, the polymer separator between the anode and cathode coatings exhibits a comparably high deformability which improves thermal contact by increased contact areas. Therefore, the interfaces between the electrode layers are modelled without an interfacial thermal contact resistance in this study, i.e. the thermal contact is assumed to be ideal.

2.4.2. Jelly roll

The jelly roll consists of many unit cells in specific geometrical arrangement. Fig. 2b shows three possible geometrical shapes, ranging from cylindrical shape to approximately pure planar shape depending on the width w of the jelly roll.

The thermal conductivity of the jelly roll is modelled based on the thermal conductivity of the unit cells and their respective geometrical arrangement. Because of the multi-layer sequence, the thermal conductivity of the jelly roll is anisotropic and can thus be split into $k_{\perp}$, i.e. the thermal conductivity perpendicular to the layers, and $k_{\parallel}$, i.e. the thermal conductivity parallel to the layers.

As the heat transfer perpendicular through the stack of layers is limited by the layers with the lowest thermal conductivity, $k_{\perp}$ is significantly smaller than $k_{\parallel}$. Depending on the position within the jelly roll and the geometrical shape, the thermal conductivity of the jelly roll k_{JR} is calculated with differently weighted proportions of $k_{\perp}$ and $k_{\parallel}$. This weighting takes into account that an increasing influence of the heat transfer bypass, i.e. along the windings, is observed on the local thermal conductivity of the jelly roll close to the windings (area B) which would otherwise be close to $k_{\perp}$ (area A). This approach is described in detail in Chapter 4.1.2.

2.4.3. Cell

For the purpose of validating the theoretically determined values of the jelly roll thermal conductivity, the following thermal cell model has been built up as a basis for the subsequently described experimental set-up in Chapter 3.2.

All but the front side of the cell are thermally insulated and assumed adiabatic. The case temperature has to be measured on both the front and the back side centre of the cell case planar surfaces. The chosen asymmetric thermal boundary condition yields an asymmetric temperature profile inside the jelly roll, which is assumed to be mainly one-dimensional and is shown in Fig. 3a.

T_{front} and T_{back} are two temperature measurements that are necessary for the calculation. The specific heat source term $\dot{q}_{el}$ is released within the jelly roll, which leads to a maximum temperature T_{max} located at the position X , compare Equation (12). The

wall thickness of the hardcase is denoted as s , the depths of the hardcase and the jelly roll are labelled as D and d , respectively. Because of its complex geometry, the jelly roll is transformed into a volumetrically equivalent cuboid.

Depth and height of the cuboid resemble the values determined for the original jelly roll, compare Table C2, whereas the width is calculated such that volume equivalence to the original jelly roll is achieved. The following bypass heat flows through the cell case between the two temperature measurements have to be considered and are calculated based on Fourier's law of heat conduction [31] applied to the geometry of the case, see Appendix B for more details.

In conclusion, the effective thermal conductivity $k_{JR,exp}$ of the jelly roll can be iteratively calculated based on Equation (11), parametrised by the measured data such as the case temperatures and the electrical heat source according to Equation (12). The approach is based on the separate calculation of the two temperature profiles left and right of T_{max} at the coordinate $x = X$, which leads to the final Equations (13) and (14). Both T_{front} and T_{back} are necessary inputs to determine the two unknown quantities T_{max} and $k_{JR,exp}$ in combination with the requirement of a continuous temperature profile at $x = X$.

$$T_{max} = T_{back} + \dot{q}_{el} \cdot \frac{X^2}{2 \cdot k_{JR,exp}} \cdot \left(1 + 2 \cdot \frac{s}{X} \cdot \frac{k_{JR,exp}}{k_{case}} \right) \quad (13)$$

$$k_{JR,exp} = \left(2 \cdot \frac{k_{JR}}{\dot{q}_{el}} \cdot \frac{T_{max} - T_{front}}{(d - X)^2} - 1 \right) \cdot \frac{k_{case}}{2} \cdot \frac{d - X}{s} \quad (14)$$

3. Experiments

All experiments were carried out with a 6.8Ah prismatic hardcase cell (SAFT, MP176065) that comprises LiCoO_2 as cathode coating material and graphite as anode coating material. Both the measurement of the individual material properties taken from a disassembled cell as well as the experimental set-up of the validation case are presented in this chapter.

3.1. Material properties

One cell specimen was carefully opened and disassembled in a glovebox under protective atmosphere (argon). The geometrical data of all cell components comprising the hardcase, jelly roll and current collectors were measured. The subsequent thermal modelling of the jelly roll as described in the result chapter is based on these measurements. The geometrical data of the hardcase is denoted in Appendix C, compare Table C1. The geometrical data of the jelly roll and its disassembled components are denoted in

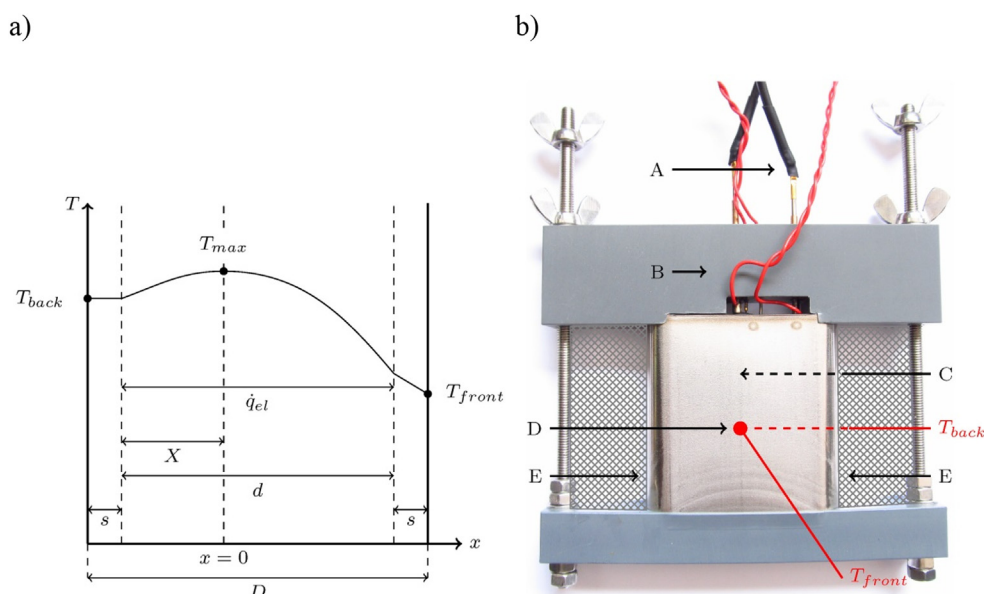


Fig. 3. a) Schematic one-dimensional temperature profile between T_{front} and T_{back} based on the asymmetric thermal boundary conditions and the volume specific heat source and b) experimental cell mounting, A: spring contact probes, B: additional voltage measurements for contact resistance calculation, C: thermal insulation on backside, D: position of the centrally thermocouples on both lateral surfaces, E: schematically thermal insulation on both curved surfaces.

Table C2 and Table C3, respectively.

The disassembled components were used to obtain the individual material properties. The electrodes were washed using ethyl methyl carbonate (EMC) solvent and small samples were punched out for characterisation after drying for 24 h at 60 °C. The temperature dependent thermal diffusivity of the current collectors and the electrode coatings has been measured by means of photothermal deflection spectroscopy (PDS) in a temperature range from 5 °C to 45 °C within a CO₂ atmosphere [17]. The specific heat capacity of the materials was determined by differential scanning calorimetry (DSC) in a temperature range from 0 °C to 50 °C [18]. The temperature ranges are restricted by the limitations of the measurement devices. The gravimetric density of the materials was obtained by He-pycnometry in the case of the electrode materials and the separator, whereas a hydrostatic balance based on Archimedes' principle was used for the metallic components. The volume to mass ratio for stated densities of the pure components was used to calculate the porosity of the electrode materials. The gravimetric density, specific heat capacity and thermal conductivity of a standard electrolyte (BASF, LP50) were determined by means of oscillating U-tube (ISO 15212-1), DSC and hot-wire method (ASTM D 2717). The porosity and the thermal conductivity of the separator are taken from literature [19,23]. An overview of the jelly roll components' material properties at 25 °C is given in Table 1.

The temperature dependent thermal conductivity of each jelly roll component is shown in Fig. 4a–d. The polynomial coefficients are given in Table 2.

Table 1
Components material properties at 25 °C [17].

component	$\rho/\text{kg m}^{-3}$	$\Psi/\%$	$c_p/J \text{ kg}^{-1} \text{ K}^{-1}$	$k/\text{W m}^{-1} \text{ K}^{-1}$
cathode	4476	29.0	750.5	1.75
aluminium	2697	—	897.8	236.3 [19]
anode	2310	47.5	759.3	10.88
copper	8929	—	382.5	394.3
separator PP	989.6	40.0 [23]	1718	0.22 [19]
electrolyte	1249	—	1642	0.18
steel case	—	—	—	24.00 [19]

The metallic electrode current collectors, copper and aluminium, are shown in Fig. 4a. The organic components separator and electrolyte exhibit the lowest thermal conductivity, as shown in Fig. 4b. The solid line indicates the porous polypropylene solid material saturated with electrolyte, as it is the case inside the jelly roll. The calculation for temperatures lower than 298.15 K is based on a linear extrapolation of the electrolyte data. It has to be considered, that the allowed temperature of the investigated electrolyte without additives ranges from 253.15 K to 353.15 K, but is nevertheless a reliable basis for a temperature dependency, as other authors observed comparable thermal conductivities [35,36]. The graphs in Fig. 4c and d show the anode and cathode coating, respectively. Both solid materials exhibit a significantly higher thermal conductivity than the electrolyte saturated porous matrix, shown by the solid lines. The dotted lines in Fig. 4c and d show the original measurement data and temperature range.

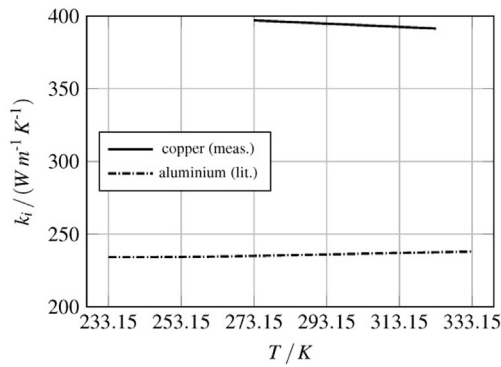
Since the PDS measurements were conducted within a CO₂ atmosphere, the adapted Krischer model (Equations (5)–(7)) is used to calculate the thermal conductivity of the solid phase first. In a second step, the effective porous media thermal conductivity in saturation with electrolyte is calculated and shown in Fig. 4b–d. The thermal conductivity of CO₂ exhibits a linear dependency on temperature [37]. These linear dependencies of the thermal conductivity are extrapolated to cover the designated temperature range for the Thermal Resistance Model (TRM) from 233.15 K to 333.15 K, where necessary.

3.2. Cell characterisation

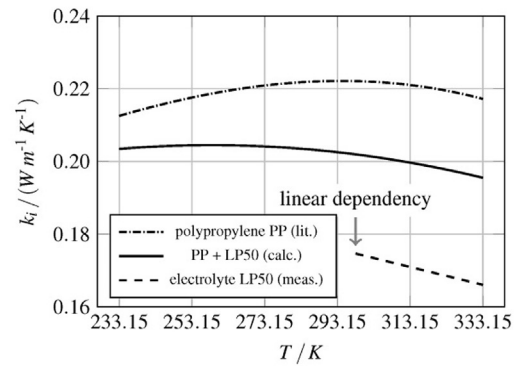
The experiments to determine the thermal conductivity perpendicular through the jelly roll were conducted with a battery test system (HPS, BaSyTec GmbH) and a digital multimeter for voltage and temperature measurements (3706 A, Keithley Instruments Inc.). The experimental set-up including the thermal insulation was placed inside a climate chamber (ARS-0680, Espec Europe GmbH) to control the ambient temperature.

For the purpose of validating the TRM, a cell mounting was designed to ensure the necessary thermal boundary conditions. A

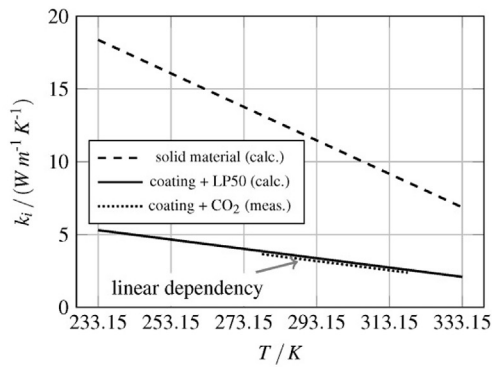
a) current collectors



b) separator and electrolyte



c) anode coating



d) cathode coating

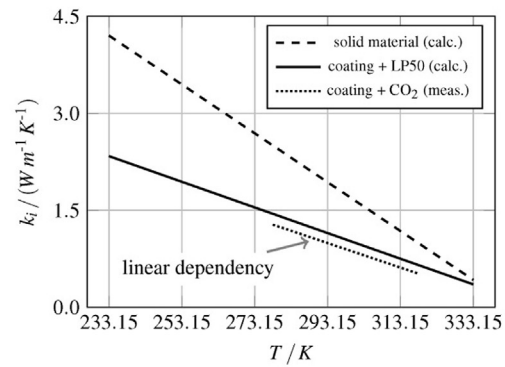


Fig. 4. Thermal conductivity k_t of the homogenised, isotropic jelly roll components: a) current collectors, b) separator and electrolyte, c) anode coating and d) cathode coating based on [17], except aluminium and separator data.

Table 2

Polynomial coefficients for temperature (K) dependent thermal conductivity k_t of the jelly roll components.

component	A/x^2	B/x^1	C/x^0
electrolyte LP50	0	2.460×10^{-4}	0.2480
PP + LP50	-1.608×10^{-6}	8.312×10^{-4}	9.706×10^{-2}
anode solid material	0	-0.1150	45.18
anode + LP50	0	-3.206×10^{-2}	12.77
cathode solid material	0	-3.778×10^{-2}	13.01
cathode + LP50	0	-1.983×10^{-2}	6.961

four-terminal sensing with gold-plated spring contact probes as electrical connection was realized to minimize the contact resistance and thereby an additional heat source as well as disruptive heat transfers to the ambient. The heat generation induced by the remaining contact resistance was determined based on an additional voltage measurement at the current spring contact probes.

A calibrated thermocouple (type K, ES Electronic Sensor GmbH) was placed on each of the two flat lateral sides of the cell case to monitor the surface temperatures. Specific thermal boundary conditions were applied to the cell case to ensure the temperature distribution between the two flat lateral surfaces, as explained in the theory and modelling section. The two curvatures and one of the flat lateral surfaces were insulated (Armaflex/NH, Armacell GmbH) to ensure an almost adiabatic boundary condition. The cell mounting with installed electrical connections and applied

insulation is shown in Fig. 3b. An additional thermal insulation at the top and the bottom surface of the cell case is not necessary, since these parts are already insulated by the cell mounting consisting of polyvinyl chloride with a comparably low thermal conductivity and an air gap between cell case and mounting.

To obtain validation data for the TRM, heat-up experiments starting from different ambient temperatures were performed. Table 3 gives an overview on the experimental settings. A symmetric zero root mean square current signal with square shape, a frequency of 10 Hz and different current ratings (relative accuracy: $\pm 0.05\%$) was used to thermally excite the cell until the thermal steady-state was reached (absolute accuracy: ± 0.05 K). The average SOC of all experiments was 70% with a deviation of $\pm 0.01\%$ due to the fast alternating current. Thereby the thermal conductivity's dependency on SOC was dispelled.

Table 3

Experimental parameter variation.

experiment	current rating	ambient temperature
# 1	0.75 C	10 °C
# 2	1.00 C	10 °C
# 3	0.50 C	25 °C
# 4	0.75 C	25 °C
# 5	1.00 C	25 °C
# 6	0.75 C	40 °C
# 7	1.00 C	40 °C

4. Results and discussion

4.1. Thermal Resistance Model

The TRM facilitates investigation of the impact of the jelly roll geometry and its components thermal material properties on the overall thermal conductivity. The determined thermal conductivities are suitable for detailed three-dimensional thermal models based on unit cell properties $k_{\perp}$ and $k_{\parallel}$ or applicable to simplified one- or two-dimensional thermal models with homogenised jelly roll thermal conductivity k_{JR} , based on a combination of unit cell properties $k_{\perp}$ and $k_{\parallel}$.

4.1.1. Thermal conductivity of the electrode stack unit cell

The series and parallel connection of the unit cell components based on Fig. 2a leads to an overall perpendicular $k_{\perp,UC}$ and an in-plane $k_{\parallel,UC}$ thermal conductivity. The calculated temperature dependencies according to Equations (2) and (3) are shown in Fig. 5a) and the polynomial coefficients are given in Table 4.

The in-plane thermal conductivity exceeds the perpendicular thermal conductivity by about one magnitude due to the current collector's high thermal conductivity. The series connection in the case of $k_{\perp,UC}$ is limited by the low thermal conductivity of the electrolyte soaked separator. The characteristic temperature dependency of $k_{\perp,UC}$ is mainly influenced by the separator. The parallel thermal conductivity $k_{\parallel,UC}$ is not limited by the lower thermal conductivities and exhibits a linear dependency due to the material properties of the current collectors and the electrode coatings. As a consequence for thermal modelling, the anisotropic ratio $k_{\perp,UC}/k_{\parallel,UC}$ is also dependant on temperature. In case of the investigated cell, it ranges between 18 and 38 for the temperature span of -40°C to 60°C . As the parallel thermal conductivity $k_{\parallel,UC}$ is also assigned to the windings of the jelly roll, the geometry has an additional impact on the overall perpendicular thermal conductivity, resembling a thermal bypass to the stacked part. As a consequence for one- or two-dimensional jelly roll thermal models without a replication of the jelly roll windings, a gain factor for the perpendicular thermal conductivity $k_{\perp,UC}$ has to be considered. The gain factor is dependent on temperature and the width to thickness ratio of the jelly roll geometry and can be calculated as a space-resolved function for $k_{\perp}(y)$ or as an averaged scalar. The following section describes the transition from unit cell to jelly roll modelling, considering the impact of geometry as mentioned before.

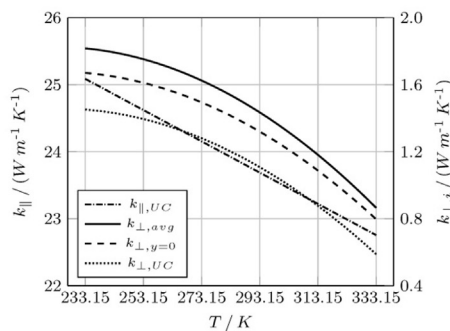


Table 4

Polynomial coefficients for temperature (K) dependent perpendicular and parallel thermal conductivity of the unit cell.

direction	A/x^2	B/x^1	C/x^0
$k_{\perp,UC}$	-7.261×10^{-5}	3.219×10^{-2}	-1.906
$k_{\parallel,UC}$	0	3.504×10^{-2}	-2.075

4.1.2. Thermal conductivity of the jelly roll

For common cooling or heating scenarios of prismatic cells via bottom or intermediate plates, the internal heat transfer towards the two planar lateral surfaces is the predominant path [3,4,18]. Most hardcase cells do not exhibit a significant thermal contact between the jelly roll and both the case bottom as well as the two narrow surfaces of the case. Fig. 6a illustrates the principal heat transfer paths and the associated thermal contact areas between the depicted jelly roll and a surrounding case.

Following this structural condition, the thermal modelling of the jelly roll focuses on the perpendicular heat transfer path. Two major contributions can be identified, that yield the overall thermal conductivity gain factor f_{TRM} of the jelly roll compared to a simple stack. The correlation is described as follows:

$$f_{TRM} = f_{geom} \cdot f_{int} \quad (15)$$

- f_{geom} the influence of the windings on the overall thermal conductivity considering solely the centre width position $y = 0$ of the jelly roll.
- f_{int} the integration of f_{geom} along the y -coordinate to cover the whole jelly roll planar section yields a further increase of the jelly roll thermal conductivity compared to a simple electrode stack.

The perpendicular thermal conductivity of the jelly roll considers additional heat flow paths over the windings, which is expressed in terms of thermal resistances in Fig. 6b.

The designated y -coordinate has to be varied in order to cover the influence of the windings. The modelling of the perpendicular thermal resistance at each y -coordinate is based on a parallel connection of three thermal resistances:

- R_l series resistance perpendicular through the layers which is based on $k_{\perp,UC}$ and therefore independent of y .

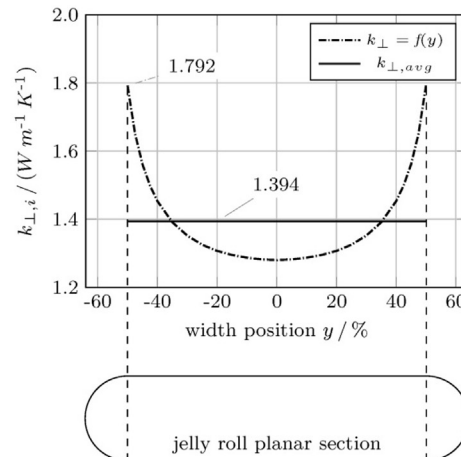


Fig. 5. a) Temperature dependent parallel and perpendicular thermal conductivity of the UC, the TRM considering the gain factor f_{geom} by $k_{\perp,avg}$ and b) the spatial dependency of jelly roll $k_{\perp}$ and the average value in the planar section of the jelly roll at 25°C temperature.

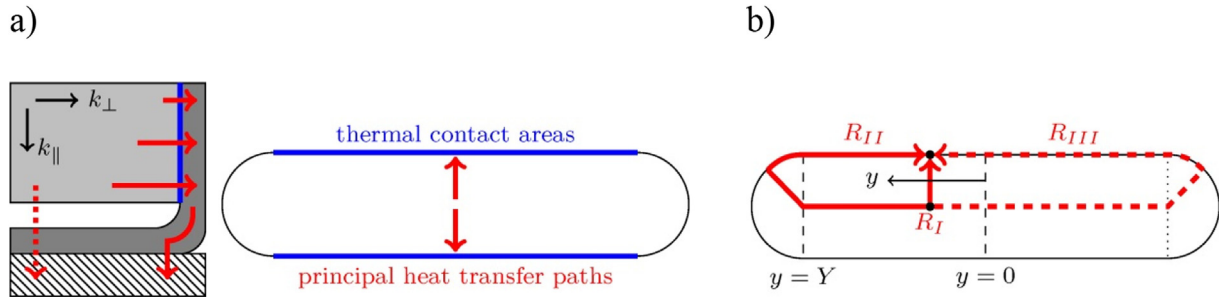


Fig. 6. a) Predominant heat transfer paths for symmetric cell temperature control via bottom or intermediate cooling plates and b) the variable thermal resistance configuration dependent on considered y-coordinate with centrically symmetry at $y = 0$.

- R_{II} series configuration of two parallel resistances along the layers of the unit cell (based on $k_{\parallel,UC}$ with a characteristic length $l_c = Y - y$) and one series resistance perpendicular through the layers in both cylindrical sections (based on $k_{\perp,UC}$).
- R_{III} resistance configuration equivalent to R_{II} but with a different characteristic length ($l_c = Y + y$) for the two parallel resistances along the layers to cover the whole planar section width of $2Y$.

The variation of the designated y-coordinate leads to a perpendicular thermal conductivity distribution that exhibits a symmetric parabolic increase towards the two cylindrical sections at the borders of the jelly roll, see Fig. 5b. The integration yields an averaged perpendicular thermal conductivity that is suitable for one-dimensional jelly roll models as well as for two- or three-dimensional models that base upon rectangular jelly roll replications. The $k_{\perp,avg}$ to $k_{\perp,y=0}$ ratio leads to the gain factor $f_{int} = 1.09$ for the designated cell geometry.

As a final step, the TRM enables the calculation of the perpendicular thermal conductivity of the jelly roll considering the temperature dependency of the underlying material properties, shown in Fig. 5a, whereas the parallel thermal conductivity is not altered by the TRM. The corresponding polynomial coefficients are given in Tables 4 and 5.

Simplified one- or two-dimensional models only depend on an appropriate perpendicular thermal conductivity, compare Fig. 6a. The ratio between the perpendicular thermal conductivity of the unit cell $k_{\perp,UC}$ and the jelly roll thermal conductivity $k_{\perp,avg}$ yields a gain factor of $f_{TRM} = 1.30$ for the designated cell geometry. The magnitude of both gain factors f_{geom} and f_{int} is eventually dependent on the aspect ratio of the designated battery cell, compare Fig. 2b. As the unit cell parameters $k_{\parallel,UC}$ and $k_{\perp,UC}$ form the calculation basis for f_{geom} and f_{int} , their temperature dependency also influences both gain factors. In this case, f_{geom} and f_{int} have only been calculated for a constant temperature of 25 °C. Based on the aspect ratio, two limiting cases can be identified:

- C_I planar stacked electrodes, pouch-bag cells
- C_{II} wound electrode geometry, cylindrical cells

The overall gain factor for very flat jelly rolls leads to $f_{TRM} = 1$ for case C_I , due to the absence of windings. In the opposite case C_{II} , the

gain factor increases significantly for jelly rolls close to a cylindrical geometry.

4.1.3. Comparison with literature data

To verify the applicability of the adapted Krischer model for effective porous media modelling, a comparison with the study of Vadakkepatt et al. [20] can be made. Herein, the authors present the normalised effective thermal conductivity k_{eff}/k_C in dependency of the cathode volume fraction. The thermal conductivity ratio of the non-active material and the active cathode particle phase $k_{NA}/k_C = 0.11$ is in accordance with our own measurements for the cathode components. A cathode porosity of $\Psi = 29\%$, compare Table 1, yields $k_{eff}/k_C \approx 0.57$. In combination with our cathode measurement $k_{cathode} = 1.56 \text{ W m}^{-1} \text{ K}^{-1}$, compare Fig. 4, that results in $k_{eff} = 0.89 \text{ W m}^{-1} \text{ K}^{-1}$. The PDS measurement of the effective cathode coating at 25 °C revealed $k_{eff,PDS} = 0.87 \text{ W m}^{-1} \text{ K}^{-1}$, which is only 2% lower than the obtained value based on the model of Vadakkepatt et al.

To illustrate the influence of the jelly roll shape on the gain factor, a connection to the paper of Forgez et al. shall be made. They inserted a thermocouple into the centre of a 26650 type cylindrical cell (LiFePO₄ cathode) for internal temperature measurement. In combination with an external sensor on the case surface, they obtained an overall thermal resistance throughout both the jelly roll and the casing at three different temperatures, resulting in an average magnitude of $R_{\perp,avg} = 3.286 \text{ K W}^{-1}$ for an average temperature of 42.5 °C [38]. Considering the jelly roll geometry and the simplifications of the thermal equivalent circuit model in terms of the non-volumetric spread heat source, their thermal resistance corresponds to an average thermal conductivity of $k_{\perp,avg} = 1.933 \text{ W m}^{-1} \text{ K}^{-1}$. This is in the same magnitude as the calculated value by the TRM at the borders of the jelly roll planar section of $k_{\perp,y=Y} = 1.792 \text{ W m}^{-1} \text{ K}^{-1}$ for 25 °C and $k_{\perp,y=Y} = 1.080 \text{ W m}^{-1} \text{ K}^{-1}$ for 42.5 °C, respectively.

An additional TRM variation of the jelly roll planar section width ($w = 0$, cylindrical jelly roll) yields a perpendicular thermal conductivity of $k_{\perp,TRM} = 2.069 \text{ W m}^{-1} \text{ K}^{-1}$ at 42.5 °C. The remaining difference of about 7% between the TRM calculation and the experimentally obtained values of Forgez et al. might be explained e.g. with a different cathode coating material. This is in accordance with the deviations of our own experimentally determined values shown in Fig. 8b.

An overview of further electrode configuration thermal conductivity values denoted in literature is given in Table 6. The literature values cover a wide range in terms of $k_{\perp}$ and $k_{\parallel}$. This can be attributed to the variety of geometry types, electrode coating materials, measurement techniques and the temperature levels, at which the values have been obtained. Furthermore, the lower pressure on the electrode stack compared to hardcase cells may increase the thermal contact resistances between the individual

Table 5
Polynomial coefficients for temperature (K) dependent thermal conductivity $k_{\perp,i}$ calculated via TRM.

value	A/x ²	B/x ¹	C/x ⁰
$k_{\perp,avg}$	-7.869×10^{-5}	3.504×10^{-2}	-2.075
$k_{\perp,y=0}$	-7.227×10^{-5}	3.219×10^{-2}	-1.906

Table 6

Literature values of electrode configuration or cell thermal conductivity.

authors	geometry	coatings	specimen	$k_{ }/\text{W m}^{-1} \text{K}^{-1}$	$k_{\perp}/\text{W m}^{-1} \text{K}^{-1}$
Bazinski et al. [16]	pouch	C/LiFePO ₄	cell	n/a	0.30–0.37
	pouch	C/Ni _x Mn _y Co _z	cell	n/a	0.41–0.45
Drake et al. [14]	26650	n/a/LiFePO ₄	cell	32.0 ± 1.6	0.15 ± 0.01
	18650	n/a/LiFePO ₄	cell	30.4 ± 1.5	0.20 ± 0.01
Fleckenstein et al. [10]	cylindrical	C/LiFePO ₄	jelly roll	n/a	0.35
Maleki et al. [8]	18650	C/LiCoO ₂	electrode stack	20.06–28.05	3.39–3.40
Maleki et al. [13]	pouch	C/LiCoO ₂	cell	32.31–36.96	1.19–1.63
Murashko et al. [12]	pouch	Li ₄ Ti ₅ O ₁₂ /LiMn ₂ O ₄	cell	n/a	0.594–0.735
Zhang et al. [15]	pouch	C/Ni _x Mn _y Co _z	electrode stack	21	0.48

layers in the case of the pouch-bag cells. The thermal conductivities $k_{\perp}$ and $k_{||}$ of a LiCoO₂ cathode specimen obtained by Maleki et al. [8,13] are in the same range as our TRM results. Especially $k_{\perp}$ of the complete pouch-bag cell ranging from $1.19 \text{ W m}^{-1} \text{K}^{-1}$ to $1.63 \text{ W m}^{-1} \text{K}^{-1}$ is in very good accordance with our unit cell values.

4.1.4. Porous media tortuosity

The original Krischer model and the adapted Krischer model differ in the value of the tortuosity f_a as described in the porous media section. The tortuosity, ranging from 0.2 for the original to 0.29–0.475 for the adapted Krischer model, has a major influence on the value of the calculated two-phase thermal conductivity of each porous media such as separator and electrode coatings. Fig. 7a and b show a direct comparison of the calculated two-phase thermal conductivities for both models. The original Krischer model exhibits a significantly lower thermal conductivity in case of the anode coating whereas only slight differences can be observed for the separator and the cathode coating. The electrolyte saturated separator is a combination of two components exhibiting thermal conductivities of the same magnitude, which is valid as well for the cathode combination, compare Table 1. Therefore, the porosity of the separator, taken from literature, has a negligible small impact as well. Solely the porous anode coating is a combination of two materials with an increased difference in terms of thermal conductivity.

4.2. Experimental results

The experimental investigation of the jelly roll thermal conductivity yields a temperature dependency similar to the single materials shown in Fig. 8a. The thermal conductivity decreases

with increasing temperature. The labels # 1 to # 7 indicate the different experimental set-ups, compare Table 3. The assigned temperatures represent the mean value of the two thermocouple measurements at the cell casing, see Fig. 3b. The electrical heat source according to Equation (12) was calculated based on an equivalent circuit model, comprising a series connection of ohmic resistances in parallel to capacitors.

Slight deviations in terms of thermal conductivity can be observed for the different combinations of electrical excitations and ambient or initial temperature respectively. The remaining deviations can be attributed to different experimental boundary conditions, e.g. the air flow of the climate chamber and the calculation of the electrical heat source.

4.3. Comparison

The experimental results are used to validate the TRM of the jelly roll by a direct comparison that is shown in Fig. 8 b. The polynomial coefficients and the least square sum (LSS) of two fit curves are given in Table 7. The plot of the thermal conductivity of the unit cell is shown as comparison. The magnitude of the experimental values is scaled down by the geometry gain factor $f_{geom} = 1.20$ to enable a direct comparison without the impact of the jelly roll windings at the centre position ($y = 0$). The second gain factor f_{int} has been omitted in this case due to the positioning of the thermocouples, compare Fig. 3b.

The $k_{JR,exp}$ fit curve is modelled based on the resulting temperature dependency of the unit cell with a constant offset as fit parameter. The values of the unit cell are in very good accordance with the modelled experiments (i.e. only about $0.005 \text{ W m}^{-1} \text{K}^{-1}$ lower) All measurements are within a range of $\pm 15\%$ of the

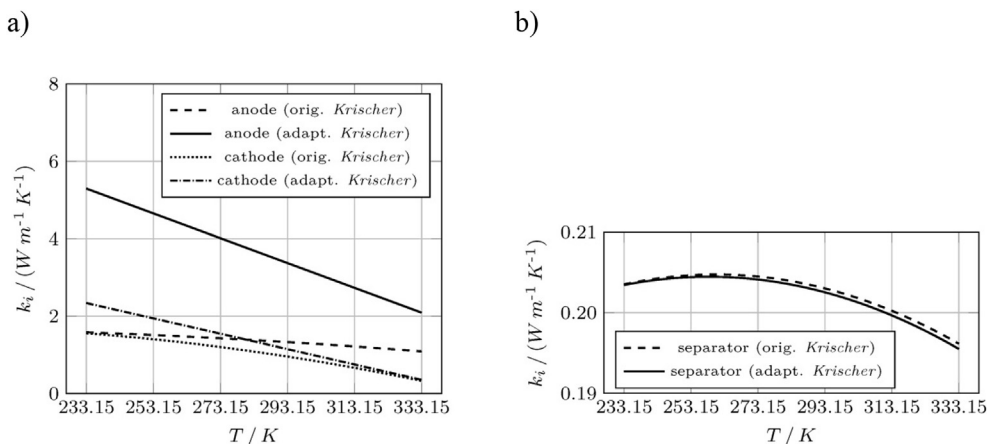


Fig. 7. Original Krischer model and its adaptation for porous media thermal conductivity modelling of a) the electrolyte saturated electrode coatings and of b) the electrolyte saturated separator.

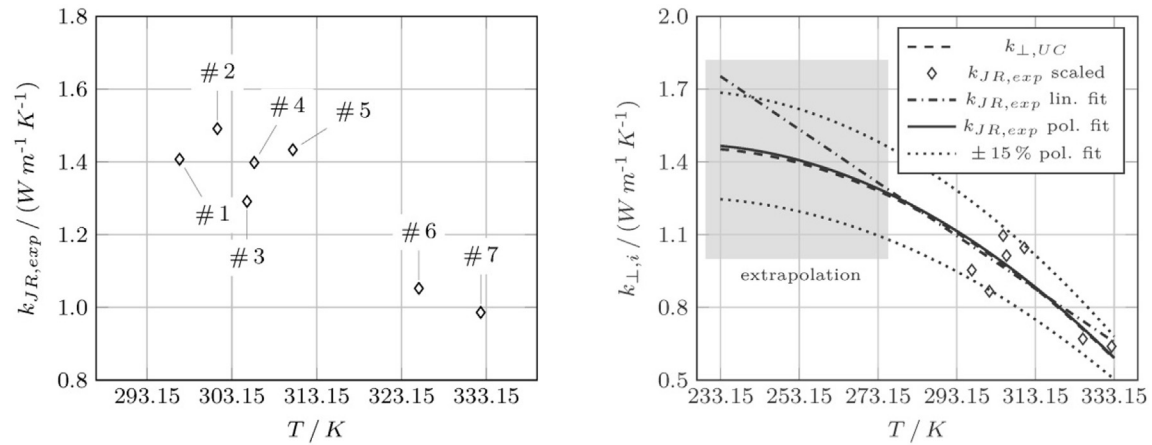


Fig. 8. a) Experimentally determined jelly roll thermal conductivity at different steady state temperatures and b) the validation of the temperature dependent thermal conductivity calculated by the TRM with scaled down experimental values.

Table 7

Polynomial coefficients for temperature (K) dependent experimentally determined jelly roll thermal conductivity scaled down for unit cell comparison.

value	A/x ²	B/x ¹	C/x ⁰	LSS
$k_{\perp,UC}$	-7.261×10^{-5}	3.248×10^{-2}	-2.174	
$k_{\perp,exp}$ lin. fit	0	-1.096×10^{-2}	4.308	0.075
$k_{\perp,exp}$ pol. fit	-7.227×10^{-5}	3.219×10^{-2}	-2.111	0.073

polynomial fit. A direct linear fit of the experimental values yields both a comparable LSS and a similar descending slope compared to the polynomial fit within the temperature range from 278.15 K to 323.15 K. The indicated extrapolation area is based on extrapolated linear dependencies of both the anode and cathode coating solid materials, copper current collector as well as the electrolyte thermal conductivity, compare Fig. 4.

5. Conclusion

A Thermal Resistance Model (TRM) was introduced as a modelling approach for the effective parallel and perpendicular thermal conductivity of electrode stacks or jelly rolls. It covers the geometrical effects of jelly rolls as well as the individual material properties such as porosity of electrode coatings and temperature dependency of the underlying parameters.

Furthermore an experimental methodology was presented, that allows the validation of the perpendicular thermal conductivity determined using the TRM. Besides the validation of modelling approach, the non-destructive experimental methodology enables the determination of the perpendicular thermal conductivity without individual material measurements of cell components. It is solely dependent on hardcase properties and does not require a complete jelly roll disassembly. The methodology can also be applied to larger geometries of prismatic hardcase cells that comprise two or more jelly rolls for instance. In contrast to other approaches described in literature [10,11], this methodology does not require a sinusoidal current source for electrical excitation since it is based on square-shaped current profiles.

It was shown, that the experimentally determined values of the perpendicular thermal conductivity and the TRM calculation are in very good agreement. The presented study successfully links experimentally determined material properties of the electrodes to the overall thermal conductivity of the corresponding full cell. According to the authors' knowledge, this was achieved for the first

time, as previous studies either dealt with the determination of single material properties or the averaged properties of full cells.

Both the TRM thermal conductivities as well as the experimentally determined values exhibit a strong dependency on temperature which is of great importance for thermal modelling. An increasing cell core temperature deteriorates the heat transfer and therefore increases the internal temperature gradient. The perpendicular thermal conductivity of the investigated hardcase cell at 60 °C is about –50% less than the value obtained at 0 °C. As an outcome of the decreasing thermal conductivity, the temperature gradient inside a cell at 60 °C will double compared to 0 °C for a constant heat flux or rather heat source. Therefore, the temperature dependent thermal conductivity has to be considered to provide safe and efficient thermal management under any operational thermal condition.

Based on the TRM calculation, the validity of simplified one- and two-dimensional thermal cell models can significantly increase by considering geometrical parts such as windings and anisotropic unit cell properties. Even complex three-dimensional thermal models with homogenised electrode stacks, jelly rolls or unit cell properties benefit from the presented in-depth unit cell layer modelling. Individual parts of the TRM, such as cathode coating material properties, can easily be replaced by other measurements for the purpose of parameter studies or the investigation of new cell configurations.

Appendix A

The indefinite integration of Equation (10) for planar geometries leads to Equation (A.1)

$$T(x) = c_0 + c_1 \cdot x - \frac{\dot{q} \cdot x^2}{2 \cdot k} \quad (A.1)$$

As a next step, c_0 and c_1 have to be determined via the boundary conditions (BC) at the border of the designated temperature distribution:

- Isothermal boundary conditions:

$$T_{x=X} = T_{BC} \quad (A.2)$$

- Heat flow through an area A:

$$\frac{dT}{dx_{x=0}} = \frac{\dot{Q}}{A} \quad (\text{A.3})$$

In case of a symmetrical or an adiabatic boundary condition, this reduces to:

$$\frac{dT}{dx_{x=0}} = 0 \quad (\text{A.4})$$

- Convection boundary conditions, e.g. an adjacent fluid with a heat transfer coefficient htc_{bc} and temperature T_{bc} :

$$\frac{dT}{dx_{x=X}} = -\frac{htc_{bc}}{k} \cdot (T(X) - T_{bc}) \quad (\text{A.5})$$

A symmetry boundary condition at the inner position $x = 0$ m leads to $c_1 = 0 \text{ K m}^{-1}$, compare Fig. 3a. The first integration constant c_0 for a convective boundary condition at the position $x = X$ results in Equation (A.6).

$$c_0 = T_{BC} + \frac{\dot{q} \cdot X}{htc_{BC}} - \frac{\dot{q} \cdot X^2}{2 \cdot k} \quad (\text{A.6})$$

Appendix B

Calculation of the bypass heat flows through the cell case between the two temperature measurements T_{back} and T_{front} :

$$\dot{Q}_{hor} = \frac{k_{case}}{l_c} \cdot A_{hor} \cdot (T_{back} - T_{front}) \quad (\text{B.1})$$

$$\dot{Q}_{ver} = \frac{k_{case}}{l_c} \cdot A_{ver} \cdot (T_{back} - T_{front}) \quad (\text{B.2})$$

The horizontal heat flow through both curvatures is labelled as $\dot{Q}_{hor}$ and $\dot{Q}_{ver}$ is the vertical heat flow through the top and the bottom part of the case. These are calculated based on the thermal conductivity k_{case} , the corresponding characteristic length l_c and cross-sectional heat flow area A . Therefore the position X of the maximum temperature T_{max} within the jelly roll, as shown in Fig. 3a, can be calculated as follows.

$$\frac{X}{d} = 2 \cdot \frac{\dot{Q}_{hor} + \dot{Q}_{ver}}{V \cdot \dot{q}_{el}} \quad (\text{B.3})$$

Table C.2

Geometrical data of the assembled jelly roll

jelly roll part	quantity	-/mm
depth (d)	—	16.41
width	—	58.07
height anode	—	53.00
height cathode (h)	—	51.00

Table C.3

Geometrical data of the jelly roll layers

layer	quantity	d/ μm
copper	34	13.00
aluminium	32	24.00
graphite	68	110.50
lithium cobalt oxide	64	95.88
separator	72	20.00

List of symbols

a	thermal diffusivity, $\text{m}^2 \text{s}^{-1}$
A	area, m^2
c_p	heat capacity, $\text{J kg}^{-1} \text{K}^{-1}$
d, D	length, m
f	gain factor
htc	heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$
h	length, m
I	current, A
k	thermal conductivity, $\text{W m}^{-1} \text{K}^{-1}$
l	length, m
m	mass, kg
ψ	volumetric porosity, %
$\dot{q}$	specific heat flow, W m^{-3}
$\dot{Q}$	heat flow, W
ρ	gravimetric density, kg m^{-3}
R_{th}	thermal resistance, K W^{-1}
s	length, m
τ	tortuosity
T	temperature, K
U	voltage, V
V	volume, m^3
w	length, m
x, X	length, m
y	length, m

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Appendix C

Table C.1

Geometrical data of the hardcase parts

hardcase part	-/mm
depth (D)	17.22
width	59.07
height	62.62
wall thickness (s)	0.38
brim thickness	0.92
brim	3.39
groove	1.06

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